

A Contextual Bandit Approach to Interactive Music Recommendation

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1 Problem Overview

We study the problem of personalized music recommendation using a reinforcement learning formulation. The task is modeled as a **contextual multi-armed bandit**, where each song corresponds to an arm, and user feedback is treated as a stochastic reward signal.

At each interaction step t , the recommender system selects a song and observes user feedback based on listening behavior such as skipping, partial listening, or explicit likes and dislikes.

2 Dataset Description

Each interaction in the dataset corresponds to a tuple:

$$(s_t, x_{t,k}, r_t)$$

where:

- s_t is the user session identifier,
- $x_{t,k} \in \mathbb{R}^d$ is the audio feature vector of song k ,
- $r_t \in \mathbb{R}$ is the observed reward.

The dataset includes Spotify audio features and implicit user feedback extracted from listening logs.

3 Context Representation

Each song k is represented by a normalized audio feature vector:

$$x_{t,k} = \begin{bmatrix} \text{danceability} \\ \text{energy} \\ \text{valence} \\ \text{tempo} \\ \text{loudness} \\ \text{acousticness} \\ \text{instrumentalness} \\ \text{speechiness} \\ \text{liveness} \end{bmatrix} \in \mathbb{R}^9$$

These features are fixed for each song and shared across all user interactions.

4 Reward Modeling

User feedback is mapped to a numerical reward as follows:

$$r_t = \begin{cases} +5 & \text{LIKE} \\ +2 & \text{FULL_LISTEN} \\ 0 & \text{PARTIAL_LISTEN} \\ -2 & \text{SKIP} \\ -5 & \text{DISLIKE} \end{cases}$$

This reward captures both explicit and implicit user preferences.

5 Personalized User Preference Model

Each user is represented by a latent preference vector:

$$\theta_u \in \mathbb{R}^9$$

The expected reward for recommending song k at time t is modeled as a linear function:

$$\mathbb{E}[r_t \mid x_{t,k}] = \theta_u^\top x_{t,k}$$

The observed reward follows a Gaussian noise model:

$$r_t = \theta_u^\top x_{t,k} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2)$$

6 Novelty-Aware Reward Extension

To account for user fatigue and repetition effects, we introduce a novelty term. Let $\Delta t_{t,k}$ denote the number of interactions since song k was last played:

$$\Delta t_{t,k} = t - t_k^{\text{last}}$$

We define the novelty function as:

$$N_{t,k} = 1 - \exp\left(-\frac{\Delta t_{t,k}}{s}\right)$$

where $s > 0$ controls the recovery speed of novelty.

The final expected reward becomes:

$$U_{t,k} = \theta_u^\top x_{t,k} \cdot N_{t,k}$$

7 Contextual Bandit Formulation

Let $\mathcal{D}_t = \{(x_{i,a_i}, r_i)\}_{i=1}^{t-1}$ be the interaction history. At time t , the recommender selects a song according to:

$$a_t = \arg \max_{k \in \mathcal{S}} Q_\alpha(U_{t,k})$$

where $Q_\alpha(\cdot)$ denotes the upper confidence bound (quantile) of the posterior reward distribution. The objective is to maximize the cumulative reward:

$$\max \sum_{t=1}^T r_t$$

8 Bayesian Linear Bandit Model

We assume a Gaussian prior over user preferences:

$$\theta_u \sim \mathcal{N}(\mu_0, \Sigma_0)$$

Given interaction history $\mathcal{D}_t$, the posterior distribution remains Gaussian and is updated using Bayesian linear regression.

This posterior is used to compute upper confidence bounds for exploration–exploitation trade-off.

9 Conclusion

This formulation provides a principled reinforcement learning framework for interactive music recommendation, leveraging contextual information, implicit feedback, and novelty modeling. The approach is fully compatible with real-world listening logs and supports online learning.